

Driver Drowsiness Detector

Hardware Project Evaluation

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1 Introduction

The aim of this project is to create a versatile piece of hardware that can be used to improve road safety. With use of a Raspberry Pi 4B and a camera (I have opted for a web cam as they are cheap and widely accessible), I will be aiming to produce a light weight AI based system to detect a drivers tiredness and aim to either wake them up, warn them of their condition or even potentially come up with a viable solution. I will further aim to inform the driver through use of a wireless digital display. For this, I am aiming to use the users own WIFI enabled device to connect to the Pi and broadcast data, warning as and other information. Since this will be presented to the user via a web page, I would further like the user to be able to interact with the system to Tailor it to themselves and their set up, to get the best function out of the project.

1.1 Why Do This?

Many accidents on the road are caused by drivers lack attention due to dreariness. With ‘10-20%’ [1] of all crashes on the road believed to have been attributed to tiredness; one in four fatalities estimated to have had tiredness play a part on UK roads, this is a serious problem. Not only this but given that an incident occurs on UK roads ‘Every 17 minutes’ [2] where ‘someone is killed or seriously injured’ [2], something needs to change.

2 Research

2.1 Tiredness Identifiers

There are several different ways to identify tiredness, especially when driving. Some examples are:

- More frequent blinking – this is because your eyes and brain function to counter fatigue and discomfort
- Harder Blinking – prolonged opening can result in them being saw and dry, so, to combat this, your body makes you blink harder and more frequently to keep them lubricated
- Delayed reaction times – neurons in your brain slow as they process information less efficiently
- Your head can begin to drop as you inevitably slowly fall asleep (worst case)
- Yawning – this is your body's attempt to both keep your brain awake as well as cool the body
- Grip loosens – your body begins to relax as you fall asleep, actions aren't made with the same intent
- Mood changes – you can often become more irritable as you fall asleep
- Snoring – if you have already fallen asleep, you relax and this can result in snoring

2.2 Driving Effectuated Factors

As you get more tired, you often lose concentration. You can't find yourself reacting more slowly and your decision making can become impaired. You can also become a more violent or unstable driver as your mood can be easily influenced as you do not have the energy to combat how you feel. Not only this, but in the worst case (in the event of falling asleep) you can no longer operate the vehicle.

2.3 Legality of Using a Mobile Device

In this project, I would like to make use of the users' own device, which will probably be their mobile phone. To get around the legal issue of using a mobile device, it will have to be securely fastened in a cradle and not held by the driver. Furthermore, I will not be able to request interaction from the driver once the vehicle is in motion unless I use hands-free [6], "for example:

- A Bluetooth headset
- Voice commands
- A dashboard holder or mat
- A windscreen mount
- A built-in satnav"

Failure to comply with this can result in 6 penalty points and a £200 fine. But this can go up to a driving ban as well as a maximum of £1000 fine.

Key Implementation Takeaways:

- **Setup Protocol:** User-interactive features must be restricted to the initial setup phase while the vehicle is stationary.
- **Motion Safety:** Integration of the device's internal accelerometer to disable interactive displays during motion, ensuring the system does not become a distraction.

2.4 Potential Models and Algorithms

2.4.1 Initial thought process – basic Convolutional Neural Netwrk (CNN):

My initial idea was to follow a methodology I had previously learnt in one of my prior modules, using a simple Convolutional model consisting of input nodes containing extracted values from an image, and then train the model to detect these trends in data to predict an outcome. For this I made use of Mediapipe's face mesh to extract features from frames extracted from videos, to form inputs into the neural network. This was my initial thought process, and with a desire to start coding, and deliver a prototype, I naively started to develop this without additional research.

This methodology and implementation is talked about in 'Simple AI Models' section of this document. Here I explain my methodology and thought process (as well as data validation) as to why I believed this method feasible in the early stages of this project.

Following this failure, which is also expressed in the 'Simple AI Models' section I expanded my research to other AI models which will follow.

2.4.2 Deep Convolutional Models:

As an expansion from the idea above, this consists of a far greater number of layers. With its main uses in image identification, this would be ideal for the task at hand. One key difference being the number of layers. This adds size and complexity to the network, which may make it more difficult to run on the Pi compared to other models due to its increased parameter size as a result of its increases in width and levels.

2.4.3 Xception Model (Extreme Inception):

A development of the normal inception model, this stands as the middle ground between regular convolutional models and depthwise separable convolutional models (depthwise + pointwise). This makes better use of parameters than the Inception v3 model rather than reducing the number of parameters. This is done through channel mixing.

By removing the multi branch capabilities of an inception model, opting for a single branch this completely decouple spatial correlations as well as branch correlations, which results in more focused learning .

To take this further, you can make use of pre-trained models. The example I have used in the Inception Models section of this paper was trained on the ImageNet dataset.

This transferred learning allows for pre-learns identification of edges, shapes and facial structure to be applied to my current data, without the need for the network to necessarily learn these itself, rather re-enforce its previous learning with my new data. This should allow for a model which learns faster, requiring less time to train, as well as a more accurate model, due to its additional pre-training. Some information gathered from [9].

2.4.4 CondenseNet:

This is an example of an “Efficient DenseNet using Learned Group Convolutions”[10]. This was designed for mobile use, with the aim of requiring limited computational power by removing connections between layers, aiming to remove redundancies which are not used in the later layers. Due to this, only one tenth of the computation is required compared to the DenseNet (a Deep Convolutional Model where all layers are connected to every other layer, rather than one layer being connected to the next). Comparable to MobileNet (an Inception model approach which focuses on efficiency and power usage rather than accuracy) but requiring less computation. This is done through pruning in the early stages of learning to form a ‘sweet spot’ [10] in between sparsity and regularity. Essentially, group (cluster) sizes are reduced (e.g. $3 \times 3 \rightarrow 1 \times 1$) which results in less computation, but a loss in accuracy, and requirement of more grouping of inputs. The idea of this is to reduce the number of connections during learning, to improve computational speeds and re-enforce routes, at the cost of accuracy.

2.4.5 Further Development Ideas / Key Takeaways:

Model Selection On such a small number of classifications and training examples for each classification, CondenseNet may not be the best option for this project. While it is said to be faster and more optimizable for a device with limited computational ability and power (like the Raspberry Pi I am using), it requires a greater quantity of training data than I have, without crossing between sources, and for the sake of this project, I believe that this may be beyond the scope of what is required for success – but this may change as the project progresses. Xception, while slower, can provide greater accuracy given the training data that I have. This would be good for experimenting on what the greatest achievable accuracy of a model given my training data could be. Not only this, but this model is available in Tflite, so it is possible for the performance to be improved while on the Pi. While not necessary, this is a nice bonus and something worth considering. Documentation on this can be found at <https://keras.io/api/applications/xception/>.

Pre-train Options: To improve the versatility of these AI models, the images that are used for training should be altered to allow for a more versatile classification. Since users will be using this at day and night, it would be beneficial to alter some of the training data (if it isn’t already at day and night) by adjusting the **brightness** to simulate darkness or lower light conditions such as changes in cloud cover or going through a tunnel. ($\pm 15^\circ$)

The camera angle and position can also change depending on how the user sets up the system. While the location of the recordings cannot be changed in post, the **angle** can. This will help to provide some fixability in the model, to still allow for detection of tiredness, even if the set-up is not perpendicular to the driver, like in most of the recordings. ($\pm 20\%$)

Another issue is **zoom**. My hardware, as well as the hardware between the different training data sources, is different. As a result, my inputs may automatically be different based on camera zoom, and how the user positions their face in the focus region, so fluctuations in zoom/crop may be a thing to consider. ($0.9 \rightarrow 1.1$ zoom/crop)

2.4.6 Information Communication:

Webpage Hosting With this project being done in python, for ease of use and versatility of existing libraries for the AI section of this project, it would be best to utilize some existing python libraries for the webpage.

Flask and Alternate Frameworks Flask is a lightweight web framework used for web-based interfaces and API's. In the case of this project, it would be used to host the local webpage. This will allow for offline communication of the drowsiness state.

Tkinter and PyQt provide a graphical use interface (idea for what I need), however, have a far more limited flexibility as well as requiring a physically connected device which will further limit how the project will function and be operated as additional wiring will create clutter as the user does not use something which is found in the vehicle anyway. As a result, neither of these frameworks are likely to be considered in the final result.

Django is a very powerful framework, however, with this comes complexity for a system with such simple routing and visualization requirements.

Another example of another framework to consider is REACT. With past experience in other projects, this has the upside that I understand it already. However, with the project being aimed to be as light as possible to improve efficiency, this client-heavy framework might not be the best for this project.

Overall, the low resource requirements and lightweight nature of this project, as well as the majority of the code base being in python, the obvious choice seems to be Flask to provide a platform-independent web interface.

API A simple way of connecting the webpage to the AI model. There are several different ways of communicating data from the tiredness detection AI to the webpage:

- **Shared File:** The idea is that both programs for AI detection and web hosting are run simultaneously with a common file to relay information between them. However, this could be more compute heavy, something I have expressed repeatedly as being something I want to reduce. Additionally, this can result in race conditions, where there is inconsistency between the file and the actual value. From a hardware point of view, with files being stored on the SD card (with limited numbers of writes) this method can result in shortfalls in the long run.
- **Shared Database:** Using some form of database (e.g. SQLite) can provide additional information as well as filtering. However, because of the simplistic nature of this project, it is likely too costly to do this. While given more time, and a greater fixation on tracking and suggested improvements to inform the user of patterns or issues, this is not in the scope of this project.
- **Use of integrated API:** This is the least expensive option in terms of computing power. You can make use of threading to keep the two lightweight, related programs,

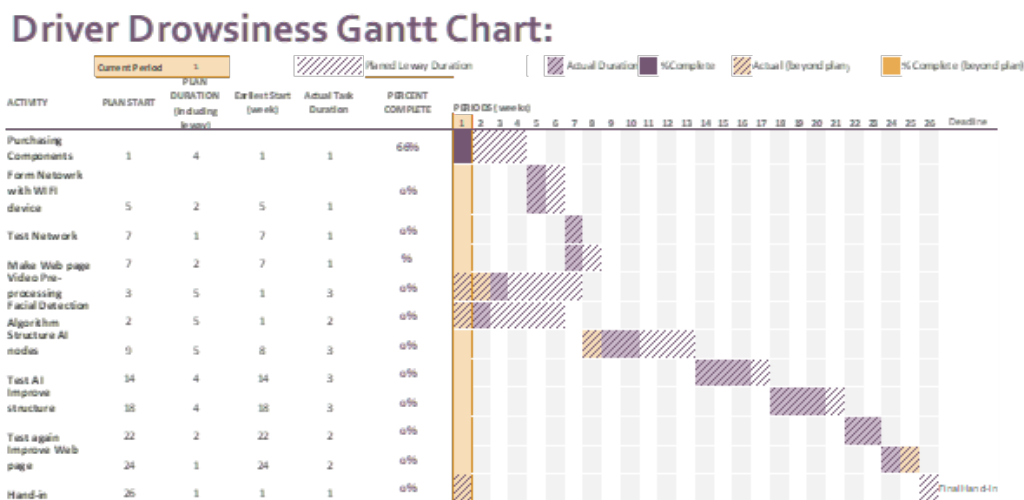
together. This makes it easier to manage and in the event of failure of the AI, the whole program knows (as opposed to being in the dark due to each program being separate).

Therefore, Option 3 (combined server and AI runner with threading) was selected as it minimizes computational overhead, avoids disk I/O bottlenecks, and keeps the architecture simple for this real-time detection application.

3 Plan

In this section, I will outline a brief outline of my intended plan of action, which will likely be reformed later on as other issues become more present during the development process.

3.1 Time Management:



Here week one refers to the starting week (w/c 20.10.2025) and week 26 being the week of the deadline (w/c 20.04.2025). The initial delay in this project is due to shipping of components. Since I do not have any of the components that I need to do this project, I will have to order them, and this takes time. However, in the meantime, I can request permission for datasets as well as gather a better understanding of what attributes can affect a driver's concentration and different potential models that I could use for my lightweight AI component.

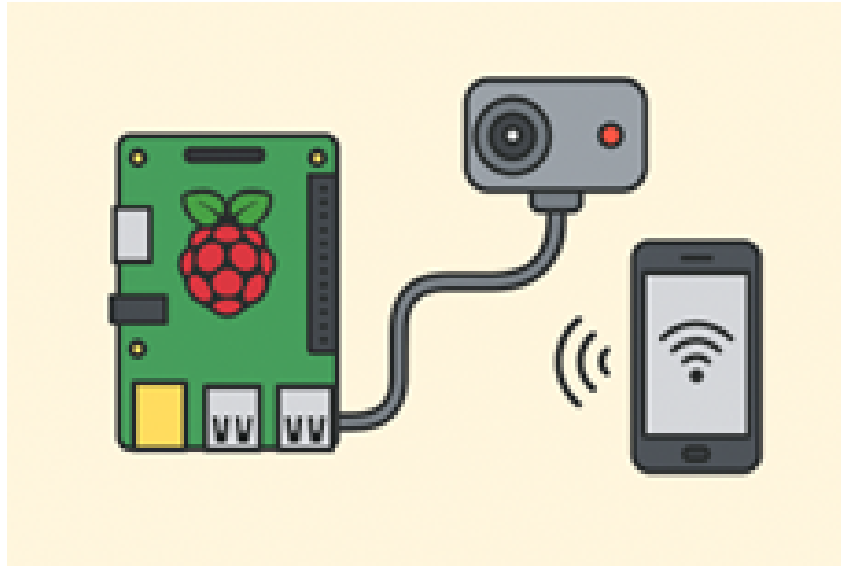
3.2 Connections:

Since raspberry pie 4 is WIFI compatible, I plan to connect the user's own device by this. Therefore, using the pi as a host device for a website that can display data and other information to the user. This is the ideal option as wired displays either require the pi to be:

- Dashboard mounted due to a small wire. This has the potential to fall or move, causing further irritation or distraction for the driver. This is against the aim of the project of keeping the driver focused on the road.
- More wires and additional screens for setting up. Once again, this takes up space in their field of vision and will create added hassle if setting up for different drivers. Most people will likely opt to not use this device, making it a futile construction.

By using the users' own wireless device, you not only remove the added expense of the screen (if released to the commercial market) but also remove clutter. Users are likely to already have phone mounts in their desired location, so why not give the idle device

some added functionality. This will also be optional, if they are already using the device for maps, it will not prevent the system from relaying information or alerting the user.



(ChatGPT generated -> prompt="please make this look nice + drawing of this")

3.3 Component Choices and Reasoning:

Raspberry Pi: For the main processing unit of this system, I have chosen a Raspberry Pi 4B. While this might be over the top in certain aspects, such as:

- Basic Graphics Processor
- 4 USB-A ports
- Ethernet capabilities
- HDMI compatibility
- GPIO Header (40-pin)
- Audio/Video jack

All of these are not necessary for this project specifically; however, I will be using this for a variety of additional projects beyond the scope of my dissertation. I chose model 4 because this has sufficient computation power to process video or images fast enough that it should not hinder the analysis. This should also be able to stay cool (while staying silent due to no fans) while running my intended AI model.

Camera: Initially, I wanted to use a rear dashcam as this was small and would provide the infra-red capabilities I desired for nighttime video, however this required additional wiring which would make this more expensive. In future, I would recommend a smaller camera module, however, for the current constraints and the plug and play capabilities, this is the superior option. These also often support 30fps. This isn't too fast of a frame rate that you are unable to extract and process the data between the frames, as well as not being too slow that you miss vital things such as blinking (which can further be used to detect tiredness).

WIFI Capable Device: This is the area of the project that can have the most flexibility in requirements. I will be making use of several different devices that I own (old phones or tablets), with the intent of users being able to use whatever they have available.

Structure: This project can be broken down into the following areas of consideration:

- **Hardware:** Selection and configuration to allow the support of a real-time driver drowsiness detection system. This includes the use of a Raspberry Pi as the main processing unit, as well as peripheral devices to capture user data required for operation.
- **Data Pre-processing:** To ensure data quality and consistency, before training the models. The handling of missing or corrupted data should be performed, as well as general normalization and clear labeling to allow for an easy workflow.
- **Image Procesing:** This is a core component of the system, enabling facial detection and analysis from still images (frames). This will consist of the following:
 - Facial Feature Extraction
 - Model Development
 - Training
 - Testing

Video Processing: Following the development of single frame processing, it is a good idea to apply this to a series of frames taken from a video input. This will allow for it to give the real-time analysis that is required. This stage will consist of:

- Extracting frames for reference
- Facial Feature Extraction
- Model development
- Training
- Testing

Following the development of the AI model, the next stage of the project is to display these findings and apply warning or other safety measures to the user.

Networking: Connecting to the user's device. Providing a connection to host a webpage on, and later allow for alerts and information to be passed

Webpage: Develop a simple webpage to display system outputs and status information.

Hosting: Host the webpage locally on the Raspberry Pi in a display-only configuration. This will help to ensure that the user does not interact with the system from within the vehicle. This will help to ensure safe operation in a vehicle setting.

Mobile Device Connectivity: Connect the user mobile device to the Raspberry Pi via local network connection to access the hosted webpage

Additional: To ensure safety and conform to laws, disable actions from the webpage (if any are sued) while in motion.

4 Datasets

For this project, I have used several different data sets from varying sources around the globe in order to try and train a more versatile system. From Turkey to China, all of these sources are publicly available and will be noted, referenced and explained in this section. Please note that additional references will be made through this document. I will outline where the datasets can be found, steps to access it for yourself as well as legal outlines that I must abide by in order to use them.

4.1 SIVAS-UNIVERSITY-OF-SCIENCE-AND-TECHNOLOGY-DRIVER-DROWSINESS-DATASET:

Collected by a Turkish University, this data set contains 19 drivers (3 female and 16 male) who recorded themselves using their phones in a variety of positions. These are in .mp4 formats. [3] Their GitHub can be found at: <https://github.com/EsraKavalci/Sivas-University-of-Science-and-Technology-Driver-Drowsiness-Dataset>

4.2 VICOMTECH:

This organization has many different datasets available such as Body pose estimation and Hands-on-wheel detection. Following my research into fatigue identifiers. I first found out about this dataset through FatigueView [5] where it was referenced as an external data set. This dataset once again has a variety of users, this time with an average age of 30. IT also outlines additional features to consider, such as glasses wearers and other driving activities such as drinking water, on the phone and other actions which may be useful to better understand how drivers act differently when fatigued. Their Page: <https://dmd.vicomtech.org/> [4]

4.3 FATIGUEVIEW:

This is not only a dataset, but also an article on how tiredness affects drivers and what identifiers exist to help identify fatigue. This uses data collected from several different people, some many different backgrounds. This data set not only contains videos of people who are already tired but also shows how people can change from awake to drowsy as time goes on. This is done through a simulated driving environment with many cameras at varying angles to help capture the most versatile data. This will be useful, as the varying camera angles will allow the model to be more forgiving to change, allowing for an individua to place the camera in the best location for them in their own car. Their article: <https://ieeexplore.ieee.org/document/9931532/figures#figures> [5]

4.4 BINA University:

A dataset collected by a university in Indonesia, the idea of the study was to “hypothesizes that drowsiness can be accurately detected in real-time through computer vision analysis” [7]. This was collected using a Raspberry Pi 5 and a camera module, taking frames at a rate of 30 FPS. This dataset takes more of a face on positioning. This will allow for more accurate extraction of features for testing as turning of the head is less likely to result in the loss of information (where other datasets with more of a bias to one side

will lose facially features when they turn away – the same change has a more dramatic effect). Furthermore, this dataset set also contains night recordings. This is important as you are:

- likely to be more tired at night
- contrasts in images and the visibility of attributes is likely to change (due to reduced lighting) so it is important to take this into account

<https://data.mendeley.com/datasets/chvz7vh2dc/1> [7]

4.5 DATASET POSITIVES AND NEGATIVES:

Camera Location: By using a variety of different camera angles, when training the model, it will be more lenient to change. Fixed locations can result in the data being more tailored, and so the AI model will be more tailored. This can influence reliability as it can result in the model being more specialized. Changes to location could potentially have a large impact on the output and its reliability. Varying countries can also have the same effect. Different countries drive on different sides of the road and therefore drive on different sides of the car.

Varying User Groups: Like location, varying users is key. Different ethnicities and languages can vastly improve data reliability. While some individuals are more expressive and others harder to identify features, this can help improve the versatility of the extracted data. While not something I would have initially considered, language spoken can play an effective role. Some languages are more articulatory and expressive so require more facial movements, which may be construed as yawning or signs of tiredness (especially if audio isn't considered).

Varying Hardware: While my project intends to use consistent hardware throughout, the variation in hardware used during data collection across different datasets provides a degree of flexibility. Since my hardware doesn't exactly match that of one or more datasets, having access to multiple datasets helps mitigate potential inaccuracies that could arise when testing on fixed data versus real-world usage scenarios.

varying Time: With some data sets spanning both night and day, there will be ranges in clarity and lighting. It can be a safe assumption that people (in most instances) are more tired at night; be this due to a long day or your circadian rhythm (your body's internal clock). This may also vary the number of 'Tired' events in a given period, simply due to time of capture.

Bais: With less women taking part in the study (across all datasets) it is likely that it will be less accurate at detecting their tiredness. While the general indicators are the same, additional features such as longer hair which can distort imaging may play a part. Additionally, glasses (while represented in the dataset) might influence facial detection and lead to false results. This is something I will have to consider later. Other areas such as eye shape and size varying through individuals might make certain groups of people more difficult to identify tiredness, while others are more prone to different patterns for being tired. Time of day will also be a factor. It is easier to see people in the day (when

they are more likely to not be tired). As a result, determining tiredness might not be as straight forward as the less clear data (relating most to the area I want to identify and train the model for) will be harder to extract information for. Not only this, with some datasets being collected from nighttime workers, they might be tired during the day, which will reduce the effectiveness of time within the data.

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